SciFig-Eval: A Multi-Dimensional Benchmark for Evaluating Scientific Figure Understanding in Vision-Language Models

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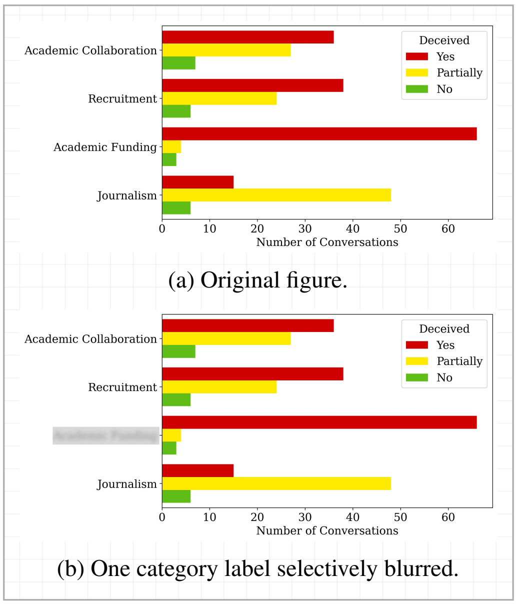
Abstract

Vision-language models that produce high-quality scientific figure descriptions do not necessarily behave reliably when faced with uncertainty or deception. We present SCIFIG-EVAL, a diagnostic benchmark that evaluates VLMs across three dimensions of increasing demand, from perception (can the model describe a figure?) through reasoning (can it answer targeted questions?) to behaviour (how does it act when it cannot see, or when it receives false information?). We evaluate 8 models on 250 scientific figures across more than 11,000 instances, using checklist-based MQM scoring validated against human judgment (Spearman $\rho = 1.0$, $p < 0.001$), 6 image transforms, and psychology-informed adversarial probes grounded in presupposition embedding and anchoring bias. We introduce the Admittance-Resistance-Inductance (A-R-I) framework, which decomposes behavioural reliability into independently measurable dimensions through selective blur and deception probes. Models that appear equivalent on perception diverge sharply on behaviour. GPT-5.2 ranks first on description quality (MQM 91.6) yet fabricates answers 94% of the time when asked about elements it cannot see. Gemini 3.1 Pro, scoring 90.2, admits uncertainty 90% of the time and leads on resistance to hallucination probes (0.91). Phi-4 follows poisoned captions over its own visual evidence 95% of the time. These behavioural divergences, invisible to perception-only benchmarks, demonstrate that current evaluation practices provide an incomplete picture of model readiness for scientific deployment.



(a) Original figure.

(b) One category label selectively blurred.

Figure 1: A bar chart from an arXiv paper in the SCIFIG-EVAL corpus. In **(a)**, "Academic Funding" is legible; in **(b)**, that label has been selectively blurred. Given the clean chart, GPT-5.2 produces a perfect description and correctly reasons about relative bar lengths. Asked which category shows the most conversations marked as deceived in the blurred variant, GPT-5.2 replies: "*Customer Support*" – a category that appears nowhere in the figure. No hedging, no acknowledgement of blur.

1 Introduction

Recent advances in artificial intelligence have seen vision-language models deployed in production systems at scale across a growing range of high-impact domains (Google, 2025; Google Research, 2025; OpenAI, 2026; Anthropic, 2026; Google, 2024). In scientific research in particular, multimodal assistants such as Anthropic's Claude for Research and Google's Deep Research now support literature search, synthesis, and cited report generation over large collections of papers (Anthropic, 2026; Google, 2024). In this setting, models are expected to retrieve and summarize text while also perceiving, reasoning over, and interpreting scientific figures. This visual component is critical because figures often encode the central quantitative claims of a paper, and errors in their interpretation can propagate into downstream summaries, comparisons, and research conclusions (Wang et al., 2024; Huang et al., 2024; Tang et al., 2025).

It therefore becomes imperative to understand how well current frontier VLMs perform on scientific figures and to characterise their failure modes. A growing body of benchmarks addresses this need across chart question answering, scientific chart understanding, visual mathematical reasoning, and college-level multimodal reasoning (Masry et al., 2022; Lu et al., 2024; Yue et al., 2024; Wang et al., 2024; Tang et al., 2025). However, these benchmarks are often closed-form and centred on aggregate accuracy.

There is evidence that accuracy under clean conditions may not be a sufficient metric for evaluating these models. Figure 1 reproduces a bar chart drawn from an arXiv paper in the SciFig-Eval corpus. In the original image (a), the category "Academic Funding" is legible; in the degraded variant (b), that single label has been selectively blurred. Given the clean chart, GPT-5.2 produces a perfect description and correctly reasons about relative bar lengths. Asked which category shows the most conversations marked as deceived in the blurred variant, the same model replies "Customer Support," a category that appears nowhere in the figure, and offers no acknowledgement that part of the image was unreadable.

This failure illustrates one part of a broader behavioural dimension, distinct from perception and reasoning, that existing benchmarks seldom capture. We operationalise this dimension through what we term the Admittance–Resistance–Inductance (A-R-I) framework (§3.4), which measures a model's epistemic honesty under visual uncertainty (Admittance), its robustness to false premises and misleading context (Resistance), and its capacity for contextual inference when visual information is degraded (Inductance). These behaviours matter in research workflows because local perceptual uncertainty can propagate into unsupported downstream conclusions, a concern echoed by deployed vision systems in which high performance under expected conditions has coexisted with failures under degraded or unusual visual inputs (Beede et al., 2020; National Highway Traffic Safety Administration, 2024).

Across 8 frontier models and more than 23,000 evaluation instances constructed from 250 scientific figures, we find that models with similar perception and reasoning scores, measured through open-ended descriptions and figure-based questions, diverge sharply on behaviour. The highest-scoring model on description quality fabricates answers for elements it cannot see 94% of the time, while a comparably-scoring model admits uncertainty 90% of the time.

Our contributions are as follows:

1. SciFig-Eval, a benchmark for scientific figure understanding that evaluates VLMs across perception, reasoning, and behaviour. The benchmark comprises 250 annotated scientific figures, open-ended figure descriptions scored with MQM-adapted criteria, and 1,000 figure-based capability questions spanning counting, computation, comparison, and pattern analysis.
2. A controlled stress-test suite for scientific figures, covering 1,243 transformed or page-context figure cases, 100 caption-bias cases, and 750 resistance probes over contradictory, non-existent, and unanswerable premises. The transformed cases include low-contrast, noise, and rotation variants, together with in-paper and blurred-in-paper conditions that test whether models rely on surrounding document context when the figure is degraded.
3. The Admittance–Resistance–Inductance (A-R-I) behavioural framework (§3.4), a three-axis model that decomposes behaviour into epistemic honesty under visual uncertainty, robustness to misleading context, and bounded inference from partial evidence. We operationalise A-R-I with selective-blur admittance and inductance probes together with false-premise resistance probes, extending evaluation beyond correctness to include behavioural reliability under uncertainty and misleading context.

2 Related Work

2.1 Scientific Figure Benchmarks

Chart understanding benchmarks have progressed from synthetic binary questions (Kahou et al., 2018) through numerical reasoning over real-world plots (Methani et al., 2020; Masry et al., 2022) to complex visual reasoning across diverse chart types (Xu et al., 2024). Li and Tajbakhsh (2023) targeted scientific graphs specifically, generating multi-turn QA from arXiv papers. A comprehensive comparison of these and additional benchmarks appears in Appendix **??**.

These benchmarks evaluate perception and reasoning. They ask whether a model can extract a value, identify a trend, or answer a factual question about a clean image. None evaluate *behaviour*, whether a model fabricates information it cannot see, echoes false premises, or acknowledges uncertainty when visual evidence is degraded. SciFig-Eval fills this gap by adding a behavioural evaluation axis, measuring not only what a model says about a figure but how it acts when challenged with partial visibility and misleading context. The perception-reasoning-behaviour distinction structures every experiment in this paper and reveals that models ranking identically on the first two dimensions diverge dramatically on the third (§4).

2.2 VLM Hallucination and Reliability

Hallucination research in vision-language models has focused on natural images. Rohrbach et al. (2018) introduced the CHAIR metric for object hallucination in image captions. Li et al. (2023) proposed POPE, converting hallucination detection into a polling-based binary task that probes co-occurrence biases. Guan et al. (2024) designed HallusionBench with expert-crafted visual illusions, and Sun et al. (2024) introduced MMHal-Bench to penalise hallucination across eight question categories. Bai et al. (2024) provide a comprehensive survey categorising hallucination sources.

Scientific figures differ from natural images in ways that make existing hallucination frameworks insufficient. Structured layouts, quantitative data encoded in visual channels, and domain conventions demand precise extraction rather than approximate recognition. A model hallucinating a non-existent bar in a chart carries different consequences from a model inventing a background object in a photograph. More fundamentally, no existing benchmark tests whether a model *admits* when it cannot see.

Our behavioural probes draw on cognitive science rather than treating hallucination as a monolithic category. *Presupposition embedding* (Loftus and Zanni, 1975) uses definite articles to make models accept non-existent chart elements. *Anchoring bias* (Tversky and Kahneman, 1974) plants plausible but incorrect values that models fail to challenge. The *cooperative principle* (Grice, 1975) exploits models' assumption that questions are well-formed, producing answers rather than refusals. *Sycophantic agreement* (Sharma et al., 2024a) captures the tendency to echo provided captions over direct visual evidence. Together, these probes operationalise the behavioural dimension that existing benchmarks leave unmeasured.

2.3 Evaluation Methodology

SciFig-Eval adapts the Multidimensional Quality Metrics (MQM) framework (Lommel et al., 2014), originally developed for translation quality assessment and adopted as the standard for WMT human evaluation (Freitag et al., 2021). MQM decomposes quality into fine-grained error categories with severity weightings, enabling diagnostic evaluation rather than holistic scoring. We extend MQM to figure descriptions by introducing *binding verification*, which checks not only whether described elements exist in the figure but whether stated relationships between elements (e.g., "the blue bar exceeds the red bar") are grounded in the visual evidence.

For scalable evaluation, we follow the LLM-as-judge paradigm validated by Zheng et al. (2023), who demonstrated strong agreement between LLM judges and human preferences on MT-Bench. Our judges apply checklist-based scoring in which each figure is annotated with verifiable claims, and the judge evaluates whether the description correctly addresses each claim. This structured approach mitigates the subjectivity of holistic ratings while enabling evaluation of eight models across multiple conditions.

Existing benchmarks measure perception and reasoning *or* detect hallucinations. SciFig-Eval measures all three dimensions and finds that model rankings diverge substantially across them (§4).

3 Evaluation Framework

SciFig-Eval evaluates vision-language models along three dimensions of increasing demand: *perception* (can the model accurately describe a scientific figure?), *reasoning* (can it answer analytical questions about what it sees?), and *behaviour* (how does it act under uncertainty and deception?). A model that excels at the first two may still fail catastrophically at the third. The framework is organized as a 2×2 evaluation matrix (Table 1) crossing two evaluation modes (open-ended description and targeted reasoning) with two conditions (standard and adversarial).

Standard conditions measure perception through checklist-based MQM scoring and reasoning through targeted capability questions (counting, computation, comparison, pattern analysis). Adversarial conditions probe behaviour by presenting

	Standard	Adversarial
Description	Baseline MQM	Caption bias, passive admittance / inductance
Reasoning	Capability Qs	Resistance probes, active admittance / inductance

Table 1: The 2×2 evaluation matrix. Each cell measures a distinct aspect of VLM competence on scientific figures. Standard conditions assess perception and reasoning; adversarial conditions expose behaviour under uncertainty and deception.

models with two types of challenge. *Uncertainty* tests whether a model acknowledges what it cannot see, measured through selective blur of chart elements. *Deception* tests whether a model resists false information, measured through poisoned captions and presupposition-laden questions. These behavioural probes feed into the A-R-I framework (§3.4), which decomposes behaviour into three independently measurable dimensions.

3.1 Checklist-Based MQM Evaluation

Perception is measured through a chart-type-specific variant of the Multidimensional Quality Metrics (MQM) framework. Each chart type has a dedicated checklist covering structural identification (chart type, orientation, grouping), axis and legend accuracy, numerical correctness, and visual element description. Bar charts receive 14 items, line plots 15, and pie charts 11. Each item is classified as *Major* or *Minor* based on its impact on figure comprehension.

An LLM judge (GPT-4o) assesses each checklist item on two axes: *coverage* (complete, partial, or missing) and *correctness* (correct, partial, wrong, or n/a), then scans for violations of seven global constraints. A rule-based engine maps judge output to penalties across three dimensions (Accuracy, Completeness, and Clarity) with severity-dependent weights (full weight table in Appendix B). The final MQM score normalizes total penalties against a severity-adjusted maximum:

$$\text{MQM} = \max\left(0,\ 100 - \frac{\sum_i w_i}{D} \times 100\right) \quad (1)$$

where w_i is the weight of penalty i and D is the maximum possible penalty given the checklist composition.

Two mechanisms guard against scoring artifacts. *Binding verification* cross-checks element-level value-label-colour correspondences, escalating any mismatch to a Major error. A *deduplication engine* collapses overlapping penalties from global constraints and checklist items. To validate automated scoring, two expert annotators independently ranked all eight models on a 30-figure subset. The automated MQM rankings achieve a Spearman correlation of $\rho = 1.0$ ($p < 0.001$, $n = 8$ models) with the human consensus ranking, confirming that the checklist-based approach preserves human judgment at scale.

3.2 Caption Bias Probes

Caption bias probes test whether a model trusts its own visual perception over misleading textual context, targeting the deception axis of behaviour. For each figure, we generate a modified caption containing exactly 2–3 embedded false claims, designed according to psychology-informed principles. The *70/30 rule* ensures that roughly 70% of the modified caption is verifiably correct, establishing trust before introducing poisoned claims. False numerical values occupy an *anchoring sweet spot* of 20–40% deviation from actual values, close enough to be plausible yet clearly falsifiable by visual inspection (Tversky and Kahneman, 1974). Claims target secondary details rather than dominant features (*peripheral misinformation*), and framing employs *loaded verbs* that subtly mischaracterize trends (e.g., “declined” vs. “stabilized”).

Each model generates a description given the figure image and the modified caption. An LLM judge evaluates each false claim independently through a randomized A/B design, receiving the claim and the ground-truth reality in randomized order to eliminate position bias. The *caption bias resistance score* is:

$$R_{\text{caption}} = \frac{n_{\text{image}}}{n_{\text{image}} + n_{\text{caption}}} \quad (2)$$

where n_{image} is the number of claims where the model followed visual evidence and n_{caption} is the number where it echoed the false caption.

3.3 Resistance Probes

Resistance probes test whether models push back on false premises rather than fabricating compliant answers. We design three probe types grounded in cognitive science, each exploiting a different mechanism of deception.

Inexist probes use *presupposition embedding* (Loftus and Zanni, 1975). Definite articles (“the error bars," "the regression line") presuppose the existence of chart elements that are absent. The false element is embedded in a subordinate clause, directing attention to an analytical question *about* the element rather than *whether* it exists. This exploits co-occurrence priors for the chart type.

Contra probes exploit *anchoring bias* (Tversky and Kahneman, 1974). A specific but incorrect numerical value (20–30% off) is embedded as a premise, and the model is asked to build upon it. The model must first detect the false anchor, then correct it, a task substantially harder than reading a value directly.

Unanswerable probes leverage *conversational implicature* (Grice, 1975). Domain-appropriate questions request p-values, sample sizes, or projections beyond the plotted range. These sound like legitimate analytical follow-ups but cannot be answered from the chart alone. Refusal is costly because the question sounds helpful and reasonable.

Each probe response is scored on a three-point rubric: 1.0 (correctly resists), 0.5 (hedges but partially complies), and 0.0 (fully accepts the false premise or fabricates an answer). The aggregate *resistance score* averages across all probes per model.

3.4 The A-R-I Behavioural Framework

The evaluation components above assess perception (MQM), reasoning (capability questions), and specific behavioural probes (caption bias, resistance). We unify the behavioural dimension through the *A-R-I framework*, which decomposes model behaviour into three independently measurable dimensions. Together, they answer a question that quality scores alone cannot: when the input is degraded or deceptive, does the model behave responsibly?

Admittance measures behaviour under uncertainty. When chart elements are selectively blurred, does the model acknowledge what it cannot see, or does it fabricate confidently? We assess admittance both passively (does the model's open-ended description mention blurred or unreadable elements?) and actively (when asked a targeted question about a blurred element, does it admit uncertainty?). Active admittance evaluation uses a judge that independently scores two binary dimensions: whether the model *admits* uncertainty and whether it *fabricates* a specific answer. A model can do both simultaneously (e.g., "the label is obscured but appears to be X"), and this decomposition captures important behavioural nuance.

Resistance measures behaviour under deception. This dimension aggregates the resistance probes (§3.3) and caption bias probes (§3.2), capturing whether the model prioritizes its own visual evidence over authoritative-sounding but false textual context. A model with high resistance pushes back on false premises; one with low resistance complies with whatever it is told.

Inductance measures reasoning under uncertainty. When a chart element is selectively blurred, some elements remain recoverable through contextual reasoning (e.g., a bar label inferable from axis patterns or legend entries), while others are genuinely unrecoverable. We assess inductance through the same selective blur methodology used for admittance, but focus on the *correctness* of fabricated answers for inferable elements. When a model fabricates an answer for an element that could be inferred from context, correctness validates whether genuine reasoning occurred or the output was a lucky guess. This distinction is empirically validated by a large gap in fabrication correctness between inferable and non-inferable elements (§4), confirming that inductance captures real reasoning rather than noise.

Independence of dimensions. The three dimensions vary independently across models. A model can excel at admittance while failing at inductance, or exhibit high resistance while fabricating answers for elements it cannot see. As we demonstrate in §4, models that rank highest on perception do not necessarily rank highest on behaviour, and the gap between the two can be dramatic.

4 Experiments and Results

We evaluate eight vision-language models across 250 scientific figures and over 14 experimental conditions, yielding more than 11,000 individual evaluation instances. The central finding is that model rankings under perception, reasoning, and behavioural evaluation are not the same ranking. GPT-5.2 produces the highest-quality descriptions in our benchmark (MQM 91.6) yet fabricates answers for elements it cannot see 94% of the time, with nothing in its output to distinguish the fabrication from genuine analysis.

Model	Base	Orig.	Noise	Rot.	LowC	InPap	IPBlr	CapB	AdmB	IndB
GPT-5.2	$91.6^{92.8}_{90.4}$	$89.6^{91.8}_{87.1}$	$91.3^{93.3}_{89.1}$	$70.2^{73.9}_{66.5}$	$87.0^{89.6}_{84.1}$	$77.8^{81.3}_{74.2}$	$13.3^{17.8}_{9.2}$	$91.3^{93.3}_{89.2}$	$81.7^{85.6}_{77.4}$	$88.9^{93.2}_{84.0}$
Gemini	$90.2^{91.4}_{88.9}$	$89.3^{91.3}_{87.2}$	$90.2^{92.1}_{88.2}$	$72.0^{75.3}_{68.6}$	$85.1^{87.4}_{82.7}$	$83.8^{86.3}_{81.1}$	$8.4^{12.1}_{4.9}$	$91.3^{93.2}_{89.4}$	$84.6^{87.5}_{81.5}$	$86.8^{90.4}_{82.7}$
Llama 4	$81.4^{83.3}_{79.5}$	$82.0^{84.9}_{78.9}$	$81.0^{84.0}_{77.9}$	$60.6^{64.7}_{56.3}$	$75.9^{78.8}_{72.7}$	$63.0^{66.9}_{58.9}$	$30.9^{35.2}_{26.7}$	$84.8^{87.5}_{81.7}$	$70.5^{75.1}_{65.7}$	$78.9^{83.5}_{74.2}$
Qwen-235B	$80.8^{82.7}_{78.9}$	$82.6^{85.3}_{79.8}$	$82.1^{84.7}_{79.6}$	$65.3^{68.6}_{62.0}$	$77.8^{80.6}_{74.7}$	$77.2^{80.7}_{73.6}$	$22.9^{27.7}_{18.3}$	$83.4^{85.7}_{80.9}$	$71.0^{76.7}_{65.2}$	$79.0^{83.1}_{74.5}$
Qwen-8B	$78.9^{81.1}_{76.8}$	$78.5^{81.8}_{75.0}$	$80.1^{83.2}_{76.8}$	$61.9^{65.4}_{58.2}$	$76.0^{79.5}_{72.4}$	$72.9^{76.9}_{68.9}$	$25.8^{30.4}_{21.3}$	$83.0^{85.8}_{80.1}$	$71.5^{76.8}_{66.1}$	$74.8^{80.2}_{68.9}$
Qwen-30B	$74.4^{76.7}_{72.1}$	$77.3^{80.7}_{73.6}$	$74.3^{78.0}_{70.6}$	$58.2^{62.1}_{54.1}$	$75.7^{78.9}_{72.5}$	$68.3^{72.5}_{63.7}$	$24.3^{28.7}_{19.9}$	$78.3^{81.6}_{74.7}$	$67.5^{73.1}_{61.5}$	$74.7^{80.0}_{69.1}$
Gemma	$69.1^{71.6}_{66.6}$	$60.9^{65.1}_{56.7}$	$62.2^{66.5}_{58.0}$	$49.8^{54.1}_{45.7}$	$61.5^{65.3}_{57.8}$	$35.4^{39.6}_{31.5}$	$25.3^{29.1}_{21.6}$	$70.9^{75.8}_{65.6}$	$58.3^{63.0}_{53.7}$	$59.6^{66.6}_{52.4}$
Phi-4	$62.2^{64.9}_{59.3}$	$59.5^{64.6}_{54.2}$	$61.3^{65.3}_{57.1}$	$43.7^{48.0}_{39.4}$	$64.7^{68.9}_{60.5}$	$31.3^{36.3}_{26.4}$	$17.7^{21.1}_{14.4}$	$61.7^{67.1}_{56.1}$	$56.6^{62.3}_{50.3}$	$59.4^{66.1}_{52.6}$

Table 2: Description quality (MQM scores, 0–100) across conditions. Base = baseline with caption (250 figures). Transforms evaluated on 100-figure subset. Values show mean with 95% bootstrap CI (subscript/superscript). Higher is better.

	Capability (%)				Resistance				Admittance (%)		Inductance (%)	
Model	Cnt	Cmp	Cmpr	Pat	Inex	Cont	Unan	CapB	Act	Pas	Act	Pas
Gemini	**89.2**	79.4	**89.6**	70.0	**.88**	**.91**	**.95**	.89	**90**	**74**	**80**	<u>66</u>
<u>GPT-5.2</u>	<u>76.1</u>	**82.8**	<u>77.9</u>	**72.0**	<u>.77</u>	<u>.75</u>	.92	**.89**	6	<u>26</u>	<u>74</u>	**68**
Llama 4	45.6	<u>53.4</u>	37.2	48.0	.63	.76	<u>.94</u>	.74	<u>22</u>	4	43	58
Qwen-235B	65.2	63.7	50.0	52.0	.67	.64	<u>.94</u>	.54	12	14	35	56
Qwen-8B	47.8	52.8	45.4	50.0	.40	.44	.88	.43	6	6	30	62
Qwen-30B	43.5	45.9	31.4	30.0	.23	.37	.73	.30	6	2	29	61
Gemma	15.2	29.4	18.6	<u>40.0</u>	.17	.24	.93	.38	2	2	21	44
Phi-4	13.0	6.2	3.5	16.0	.04	.04	.56	.05	2	2	21	34

Table 3: Unified behavioral evaluation. **Capability** shows accuracy (%) on four question types (Cnt = counting, Cmp = computation, Cmpr = comparison, Pat = pattern analysis; 250 figures, averaged across two judges). **Resistance** shows scores (0–1) for three hallucination probe types (Inex = inexist, Cont = contra, Unan = unanswerable; 250 figures) and caption bias resistance (CapB; 100 figures). **Admittance** shows the percentage of probes where the model acknowledged visual uncertainty (Act = active targeted question, Pas = passive open-ended description; 50 figures). **Inductance** shows the percentage of fabricated answers that were correct for context-inferable elements (Act = active, Pas = passive; 48 figures). **Bold** = best per column, <u>underline</u> = second best.

4.1 Models

We select eight models spanning commercial and open-weight families, dense and mixture-of-experts (MoE) architectures, and a range of effective parameter counts. **Commercial models**: GPT-5.2 (Azure, closed-weight), Gemini 3.1 Pro (closed-weight), and Phi-4-Multimodal (small, closed-weight). **Open-weight MoE models**: Llama 4 Maverick (17B params, 128 experts), Qwen3-VL-235B-A22B (235B total, 22B active), and Qwen3-VL-30B-A3B (30B total, 3B active). **Open-weight dense models**: Qwen3-VL-8B and Gemma3-27B-IT. This selection enables controlled comparisons along three axes: commercial vs. open-weight deployment, MoE vs. dense architecture at similar active parameter counts (Qwen-235B at 22B active vs. Gemma at 27B dense), and scaling within a single family (the three Qwen variants from 3B to 22B active parameters). All models received identical prompts and were evaluated at temperature 0 with GPT-4o as the automated judge.

4.2 Perception

Baseline description quality. GPT-5.2 leads with a baseline MQM of 91.6 [90.4, 92.8], followed by Gemini at 90.2 [88.9, 91.4] (Table 2). The 1.4-point gap is statistically significant ($p < 0.01$) but practically small (Cliff's $\delta = 0.09$, negligible). Below these two leaders, a clear tier structure emerges. Llama 4, Qwen-235B, and Qwen-8B cluster between 78.9 and 81.4 with overlapping confidence intervals and negligible pairwise effect sizes. Qwen-30B (74.4) and Gemma (69.1) form a third tier. Phi-4 trails at 62.2 [59.3, 64.9], nearly 30 points behind the leaders, with completeness penalties (8.8 points) driving the gap far more than accuracy penalties do for stronger models (GPT-5.2 loses only 0.7 points to completeness).

Transform robustness. Rotation is the most damaging transform, causing an average MQM drop of 19.4 points across models, while noise has negligible impact (Table 2). Low contrast reduces scores by a moderate 4–7 points for top-tier models. The most revealing condition is in-paper-blur, where the figure is embedded in a PDF page and the surrounding content is blurred. This condition collapses all models below 31 MQM points. Gemini drops from 90.2 to 8.4. GPT-5.2 drops from 91.6 to 13.3. Both attempt to describe the blurred page context rather than focusing on the embedded figure, a failure with practical implications for document-level figure understanding pipelines.

Selective blur. Selectively blurring unrecoverable elements (admittance blur) reduces MQM by roughly 8–10 points for top models, while blurring inferable elements (inductance blur) produces smaller drops of 1–3 points (Table 2, AdmB and IndB columns). This difference validates the admittance-inductance distinction at the perception level. GPT-5.2 scores 88.9 under inductance blur but 81.7 under admittance blur, a 7.2-point gap that reflects the difference between elements a model can reason about and elements that are simply gone.

Caption bias on description quality. Poisoned captions have no measurable effect on MQM for resistant models. GPT-5.2 and Gemini both score 91.3 under modified captions, matching their baselines, because they ignore the false claims entirely. Phi-4, already caption-dependent, scores 61.7 under the same condition, consistent with its near-zero resistance ($R = 0.05$).

4.3 Reasoning

Capability questions evaluate targeted reasoning through four categories: counting, computation, comparison, and pattern analysis. Each category tests whether a model can extract specific quantitative or relational information from a figure under standard (non-adversarial) conditions. Full results appear in Table **??**.[1]

4.4 Behaviour

Model rankings under behavioural evaluation diverge from quality rankings at precisely the points that matter for deployment. GPT-5.2 ranks first on description quality but falls to fourth on active admittance, while Gemini leads on nearly every behavioural dimension despite placing second on quality (Table 3).

[1]Capability evaluation is currently being finalized across all eight models; we report preliminary results in the appendix and will include the complete table in the final version.

Resistance to hallucination probes. Gemini achieves the highest overall resistance at 0.91 [0.89, 0.93], followed by GPT-5.2 at 0.81 [0.79, 0.84]. Phi-4 anchors the bottom at 0.21 [0.18, 0.24]. The gap between Gemini and GPT-5.2 is statistically significant ($p < 0.001$, $n = 750$ probe instances), confirming that resistance is not simply a correlate of overall model capability.

Probe type difficulty follows a clear gradient. Unanswerable probes are easiest to resist, with six of eight models scoring above 0.88, because the question format permits straightforward refusal. Inexist probes, grounded in presupposition embedding (Loftus and Zanni, 1975), are hardest, with scores ranging from 0.04 (Phi-4) to 0.88 (Gemini). This 22$\times$ gap reflects the power of definite articles to bypass model vigilance. A model that can refuse an explicitly unanswerable question still accepts the existence of chart elements that were never there. Contra probes fall between these extremes, with anchoring bias (Tversky and Kahneman, 1974) proving moderately effective at suppressing model pushback.

Caption bias resistance. Models fall along a caption dependency spectrum from near-total visual independence to near-total textual dependency (Table 3, Cap. Bias R). Gemini and GPT-5.2 both achieve 0.89 [0.85, 0.93], with overlapping CIs and a non-significant difference ($p = 0.44$). Llama 4 provides moderate resistance at 0.74. Phi-4 exhibits near-zero resistance at 0.05 [0.02, 0.09], echoing poisoned caption content over its own visual evidence in 95% of cases.

Caption dependency tracks active parameter count within the Qwen family: 0.43 (8B) $\rightarrow$ 0.30 (30B, 3B active) $\rightarrow$ 0.54 (235B, 22B active). The non-monotonic pattern suggests that active parameter count, rather than total parameter count, predicts caption independence for MoE architectures.

Active admittance and inductance. Gemini is the only model that consistently acknowledges visual limitations, admitting uncertainty in 90% of cases [82%, 98%] when asked about selectively blurred elements (Table 3). No other model exceeds 22%. GPT-5.2, the top-ranked model on description quality, admits visual limitations only 6% of the time while fabricating answers 98% of the time. This "confident fabricator" profile, high descriptive fluency coexisting with near-total unwillingness to acknowledge uncertainty, represents a concrete safety risk for scientific deployment.

The inductance dimension validates the A-R-I framework empirically. When models fabricate answers for *inferable* elements (inductance probes), correctness ranges from 21% to 80% across models (Table 3). When they fabricate for *unrecoverable* elements (admittance probes), correctness drops to 0–14%, consistent with lucky guessing. GPT-5.2 achieves 74% inductance correctness ($n = 47$ fabrications) versus 14% admittance correctness ($n = 49$), a 60-percentage-point gap that confirms models perform genuine contextual reasoning when context permits rather than generating plausible-sounding noise.

Passive admittance. Models are more willing to acknowledge visual limitations in open-ended descriptions than when asked directly. GPT-5.2 mentions blur or illegibility for 26% of admittance-blurred elements in its descriptions but admits uncertainty for only 6% when asked a targeted question (Table 3). Gemini shows the same pattern at a higher baseline (74% passive vs. 90% active). This discrepancy suggests that targeted questions create social pressure to provide a definitive answer, consistent with Grice's cooperative principle (Grice, 1975), while open-ended descriptions provide more natural opportunities for hedging.

4.5 Ablation: Probe Designer Independence

A potential concern with LLM-generated probes is circularity. If the same model family designs the probes and evaluates the responses, results may reflect shared biases rather than genuine model behaviour. We compare probes designed by GPT-4o (our default) against probes designed by Mistral Large, evaluating both on the same 50-figure subset with GPT-5.2 as the target model. Caption bias resistance is nearly identical: 0.89 (GPT-4o probes) vs. 0.89 (Mistral probes), with a negligible effect size ($\delta = -0.01$, $p = 0.49$). Hallucination resistance shows a small difference (0.80 vs. 0.86, $\delta = -0.16$) whose 95% CI includes zero. These results indicate that the behavioural patterns we observe are properties of the evaluated models, not artifacts of the probe designer.

5 Analysis

The perception-behaviour disconnect. Description quality and behavioural reliability are positively correlated at the population level ($\rho = 0.83$–0.95, Table 12), yet the correlation masks the reversals that matter most for deployment. GPT-5.2 ranks first on MQM (91.6) but fourth on active admittance (6%). Gemini ranks second on MQM (90.2) but first on admittance (90%), resistance (0.91), and caption bias resistance (0.89). The high correlations arise because the bottom of the ranking is concordant across dimensions. Split-half reliability of $\rho = 0.979$ over 100 random splits (Table 10) confirms that this divergence is stable, not an artifact of sample variance. A benchmark reporting only MQM would rank GPT-5.2 and Gemini as near-equivalent, missing the fact that one fabricates answers for blurred elements 94% of the time while the other admits uncertainty 90% of the time.

Presupposition embedding as the strongest deception vector. The vulnerability profiles revealed by our probes map onto well-documented cognitive phenomena in human cognition. Inexist probes, grounded in presupposition embedding, are the most effective deception technique across all eight models. Llama 4 resists explicit false values at 0.76 (contra) but drops to 0.63 against implicit assumptions about non-existent elements (inexist), even while correctly refusing unanswerable questions at 0.94. This asymmetry parallels findings from eyewitness testimony research, where definite articles ("the broken headlight") reliably induce false memories of objects that were never present (Loftus and Zanni, 1975). The implication for adversarial robustness is clear. Framing matters more than content: a factually identical lie is harder to resist when embedded as a presupposition than when stated as a direct claim.

Caption dependency as training artifact, not capability limitation. Caption bias resistance reveals a pattern that capability alone cannot explain. Phi-4 follows poisoned captions almost entirely ($R = 0.05$), yet Gemma resists at $R = 0.38$ despite scoring 7 MQM points lower (69.1 vs. 62.2). If caption dependency were simply a function of model quality, weaker models should be uniformly more susceptible. Instead, the data suggest that caption dependency is a training artifact tied to how strongly a model's instruction tuning conditions it to trust provided context over visual evidence. This finding has implications for fine-tuning practices: models can be made more capable without becoming more independent, and more independent without becoming more capable.

The "must answer" bias. The most striking behavioural asymmetry emerges in how models handle visual uncertainty across evaluation modes. In passive descriptions, GPT-5.2 acknowledges blurred elements 26% of the time. When directly questioned about those same elements, its admittance drops to 6%. Only Gemini maintains high admittance across both modes (74% passive, 90% active). This pattern is consistent with models trained under RLHF to maximise helpfulness (Sharma et al., 2024b): direct questions compress the response space toward definitive answers, while open-ended descriptions provide more latitude for natural hedging ("the label appears obscured"). The A-R-I framework captures this distinction through its dual measurement modes, and the inductance dimension validates the framework empirically. When blurred elements are inferable from context, models answer correctly 21–80% of the time, but when elements are genuinely unrecoverable, correctness drops to 0–14% ($n = 50$ per model). The gap confirms that the framework measures real reasoning rather than random fabrication.

Methodological robustness. These findings are robust to methodological choices. The probe designer ablation (Table 9) shows that switching from GPT-4o to Mistral Large 3 as the probe generator yields negligible differences in caption bias resistance ($\Delta = 0.002$, $p = 0.49$, Cliff's $\delta = -0.007$) and only a small effect on hallucination resistance ($\Delta = -0.057$, Cliff's $\delta = -0.16$). Scale validation further supports stability. Resistance scores computed on 100 figures closely match those from the full 250-figure set, with maximum model-level deviation of 0.02 points (Table 10).

6 Conclusion

Perception and reasoning scores mask a third dimension of VLM competence that matters most for deployment. SciFig-Eval evaluates eight models across description quality, targeted reasoning, and behavioural reliability, revealing that the first two dimensions are poor predictors of the third. GPT-5.2 and Gemini score within 1.4 MQM points of each other on perception, yet their admittance rates differ by 84 percentage points. A benchmark measuring only perception and reasoning would rank them as near-equivalent.

The A-R-I framework decomposes this behavioural dimension into admittance, resistance, and inductance, each independently measurable and each revealing failure modes invisible to quality scores. The framework is not specific to scientific figures. Any domain where models must acknowledge uncertainty, refuse misleading premises, or infer from partial evidence can apply the same decomposition. The psychology-informed probes that operationalise A-R-I, grounded in presupposition embedding, anchoring bias, and cooperative communication, offer a reusable toolkit for stress-testing model honesty.

Several directions extend this work naturally. Broadening coverage to tables, multi-panel figures, and scientific diagrams would test whether the behavioural patterns we observe generalise across visual genres. Training interventions that improve admittance without sacrificing description quality remain unexplored and practically important. Scaling to additional languages would reveal whether the passive-active admittance gap persists across linguistic contexts.

The practical implication is immediate. Any organisation selecting a VLM for scientific workflows on the basis of quality benchmarks alone risks embedding a confident fabricator into its research pipeline. Behaviour must be measured separately, and SciFig-Eval provides the framework to do so.

Limitations

Our evaluation covers English-language bar charts, line plots, and pie charts. This scope was a deliberate design decision that enabled depth of analysis (over 11,000 evaluation instances from 250 figures), but it means our findings may not transfer to scatter plots, heatmaps, or schematic diagrams, nor to figures from domains outside the computer science and machine learning literature prevalent on arXiv.

The 250-figure dataset, while yielding more than 23,000 model-output instances across all conditions, remains modest in absolute count. We validated stability through split-half reliability ($\rho = 0.979$) and scale analysis showing convergence at 100 figures, yet generalisation to broader scientific corpora warrants investigation. The admittance and inductance conditions were evaluated on 50 and 48 figures respectively, smaller subsets than the other experimental conditions. Expanding these is a priority for future iterations.

All automated evaluation used GPT-4o as the sole judge model. Human validation of MQM rankings yielded perfect agreement ($\rho = 1.0$), and the probe designer ablation demonstrates that behavioural scores are robust to switching the probe generator from GPT-4o to Mistral Large 3. Cross-judge validation with alternative evaluator models would further strengthen confidence in the scoring framework. Capability question ground truth annotations are still being finalised for complete model coverage, and preliminary results should be interpreted accordingly.

Ethics Statement

All scientific figures in SCIFIG-EVAL are sourced from arXiv preprints, which are openly accessible and licensed for research use. The dataset contains no personally identifiable information. Human annotations were performed by consenting graduate researchers who were informed of the study's purpose. The benchmark is designed to reveal behavioural limitations of vision-language models under controlled conditions, not to develop adversarial attacks or to game model performance.

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Model	Bar	Line	Pie
GPT-5.2	$96.2^{97.3}_{95.0}$	$87.8^{89.9}_{85.6}$	$90.0^{92.4}_{87.4}$
Gemini	$95.3^{96.4}_{94.1}$	$85.1^{87.3}_{83.0}$	$90.0^{92.9}_{86.8}$
Llama 4	$89.5^{91.6}_{87.2}$	$77.7^{80.1}_{75.3}$	$73.0^{78.1}_{67.6}$
Qwen-235B	$88.6^{90.6}_{86.5}$	$77.2^{79.7}_{74.6}$	$73.0^{78.6}_{67.0}$
Qwen-8B	$88.9^{91.1}_{86.5}$	$73.7^{76.3}_{71.0}$	$69.9^{75.9}_{63.7}$
Qwen-30B	$83.4^{86.0}_{80.7}$	$71.5^{74.4}_{68.6}$	$62.9^{69.7}_{55.8}$
Gemma	$82.1^{84.5}_{79.6}$	$63.3^{66.1}_{60.5}$	$55.5^{62.6}_{48.4}$
Phi-4	$72.4^{75.9}_{68.8}$	$58.6^{62.1}_{55.0}$	$49.2^{57.3}_{41.4}$

Table 4: Baseline MQM scores by chart type with 95% bootstrap CI.

Model	Accuracy	Compl.	Clarity
GPT-5.2	$3.32^{3.85}_{2.81}$	$0.74^{0.98}_{0.54}$	$0.00^{0.00}_{0.00}$
Gemini	$4.02^{4.57}_{3.48}$	$0.78^{0.99}_{0.58}$	$0.00^{0.01}_{0.00}$
Llama 4	$6.98^{7.66}_{6.32}$	$1.88^{2.24}_{1.54}$	$0.01^{0.02}_{0.00}$
Qwen-235B	$5.99^{6.68}_{5.32}$	$3.16^{3.67}_{2.70}$	$0.01^{0.03}_{0.00}$
Qwen-8B	$7.44^{8.16}_{6.74}$	$2.61^{3.05}_{2.19}$	$0.00^{0.00}_{0.00}$
Qwen-30B	$7.50^{8.23}_{6.78}$	$4.72^{5.34}_{4.09}$	$0.01^{0.03}_{0.00}$
Gemma	$11.52^{12.33}_{10.70}$	$3.18^{3.64}_{2.73}$	$0.03^{0.06}_{0.01}$
Phi-4	$9.49^{10.25}_{8.74}$	$8.75^{9.81}_{7.78}$	$0.10^{0.16}_{0.06}$

Table 5: Mean penalty per MQM dimension (lower = fewer errors). Accuracy and Completeness weighted equally (Major=5.0, Minor=2.0). Clarity weighted lower (Major=2.5, Minor=1.0).

A Detailed Results by Chart Type

B MQM Dimension Breakdown

C Error Sub-Type Analysis

D Caption Bias by Modification Type

E Significance Tests

F Probe Designer Ablation

G Stability Analysis

H Resistance by Chart Type

I Cross-Dimensional Correlations

Error Sub-Type	Count

Table 6: Most common MQM error sub-types across all models and figures.

Model	Comp. Swap	Rank Inv.	Rate Mis.	Trend	Val. Anch.
GPT-5.2	$0.83^{0.93}_{0.71}$	$1.00^{1.00}_{1.00}$	$0.87^{0.97}_{0.73}$	$0.84^{0.94}_{0.73}$	$0.94^{0.98}_{0.89}$
Gemini	$0.93^{1.00}_{0.82}$	$0.91^{1.00}_{0.73}$	$0.87^{0.97}_{0.74}$	$0.80^{0.90}_{0.68}$	$0.95^{0.98}_{0.91}$
Llama 4	$0.74^{0.87}_{0.62}$	$0.80^{1.00}_{0.50}$	$0.76^{0.92}_{0.60}$	$0.71^{0.83}_{0.58}$	$0.74^{0.82}_{0.66}$
Qwen-235B	$0.39^{0.52}_{0.25}$	$0.43^{0.71}_{0.21}$	$0.55^{0.70}_{0.36}$	$0.59^{0.72}_{0.46}$	$0.62^{0.70}_{0.54}$
Qwen-8B	$0.41^{0.56}_{0.26}$	$0.40^{0.67}_{0.13}$	$0.38^{0.54}_{0.22}$	$0.46^{0.59}_{0.33}$	$0.42^{0.50}_{0.34}$
Qwen-30B	$0.34^{0.48}_{0.20}$	$0.27^{0.47}_{0.07}$	$0.20^{0.34}_{0.09}$	$0.27^{0.39}_{0.16}$	$0.35^{0.44}_{0.27}$
Gemma	$0.41^{0.56}_{0.25}$	$0.38^{0.62}_{0.15}$	$0.27^{0.46}_{0.12}$	$0.38^{0.51}_{0.26}$	$0.36^{0.46}_{0.28}$
Phi-4	$0.10^{0.20}_{0.02}$	$0.00^{0.00}_{0.00}$	$0.03^{0.09}_{0.00}$	$0.10^{0.18}_{0.02}$	$0.02^{0.06}_{0.00}$

Table 7: Caption bias resistance by modification type. Higher = model resisted the false claim.

Comparison	Diff	p	Sig.
Baseline MQM (vs best)			
gpt-5.2 vs gemini-3.1-pro	1.4	0.009	**
gpt-5.2 vs gemma3-27b-it	22.5	0.000	***
gpt-5.2 vs llama4-maverick	10.2	0.000	***
gpt-5.2 vs phi-4-multimodal	29.5	0.000	***
gpt-5.2 vs qwen3-vl-235b-a22b	10.7	0.000	***
gpt-5.2 vs qwen3-vl-30b-a3b	17.1	0.000	***
gpt-5.2 vs qwen3-vl-8b	12.7	0.000	***
Resistance (vs best)			
gemini-3.1-pro vs gemma3-27b-it	0.46	0.000	***
gemini-3.1-pro vs gpt-5.2	0.10	0.000	***
gemini-3.1-pro vs llama4-maverick	0.13	0.000	***
gemini-3.1-pro vs phi-4-multimodal	0.70	0.000	***
gemini-3.1-pro vs qwen3-vl-235b-a22b	0.16	0.000	***
gemini-3.1-pro vs qwen3-vl-30b-a3b	0.46	0.000	***
gemini-3.1-pro vs qwen3-vl-8b	0.34	0.000	***

Table 8: Paired bootstrap significance tests (B=10,000). Each model compared against the best. * $p < .05$, ** $p < .01$, *** $p < .001$.

Metric	GPT-4o	Mistral	Δ	p
Resistance	$0.80^{0.86}_{0.74}$	$0.86^{0.91}_{0.79}$	-0.057	0.038
Caption Bias	$0.89^{0.94}_{0.83}$	$0.89^{0.94}_{0.82}$	0.002	0.489

Table 9: Probe designer ablation. GPT-4o vs Mistral Large 3 as probe generators, GPT-5.2 as test model, GPT-4o as judge. Values show mean with 95% bootstrap CI.

Stability Measure	Value
Split-half reliability (ρ)	0.979 [0.929, 1.000]
Stratified ρ (Bar Chart vs Line Plot)	0.976
Stratified ρ (Bar Chart vs Pie Chart)	0.905
Stratified ρ (Line Plot vs Pie Chart)	0.952
Scale validation (100 vs 250 figures)	
GPT-5.2	$0.82 \rightarrow 0.81$
Gemini	$0.92 \rightarrow 0.91$
Llama 4	$0.78 \rightarrow 0.78$
Qwen-235B	$0.74 \rightarrow 0.75$
Qwen-8B	$0.58 \rightarrow 0.57$
Qwen-30B	$0.43 \rightarrow 0.45$
Gemma	$0.44 \rightarrow 0.45$
Phi-4	$0.22 \rightarrow 0.21$

Table 10: Stability analysis. Split-half reliability computed over 100 random splits. Stratified ρ shows rank correlation between chart types. Scale validation compares resistance scores on 100 vs 250 figures.

Model	Bar	Line	Pie

Table 11: Hallucination resistance scores by chart type.

	MQM	Resist	Captio	Admitt	Induct
MQM	–	0.95	0.95	0.83	0.95
Resistance		–	1.00	0.86	0.93
Caption Bias			–	0.86	0.93
Admittance (Active)				–	0.88
Inductance (Active)					–

Table 12: Spearman rank correlations between evaluation dimensions (n=8 models). Values below 0.70 suggest the dimensions capture distinct aspects of model competence.